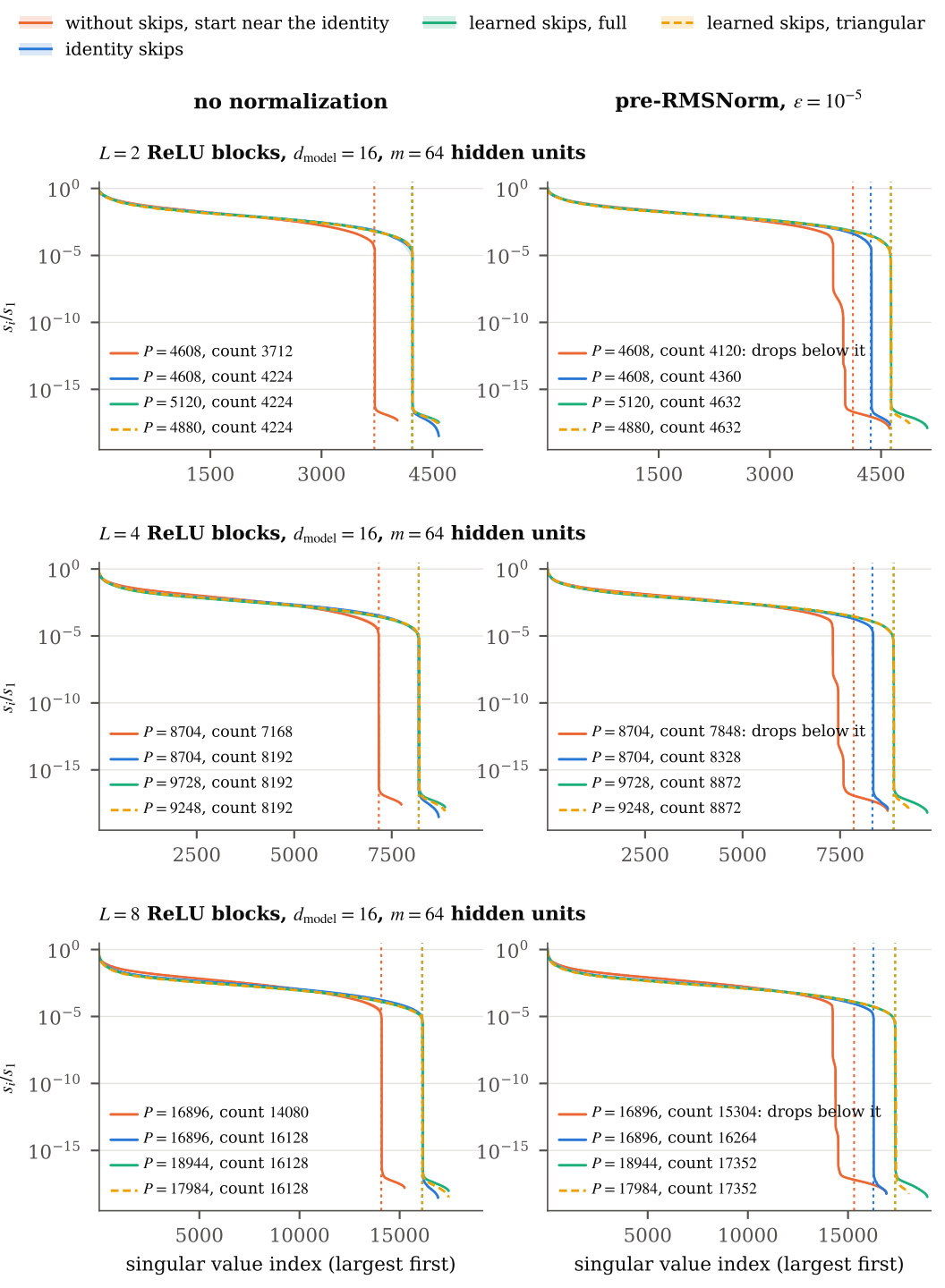
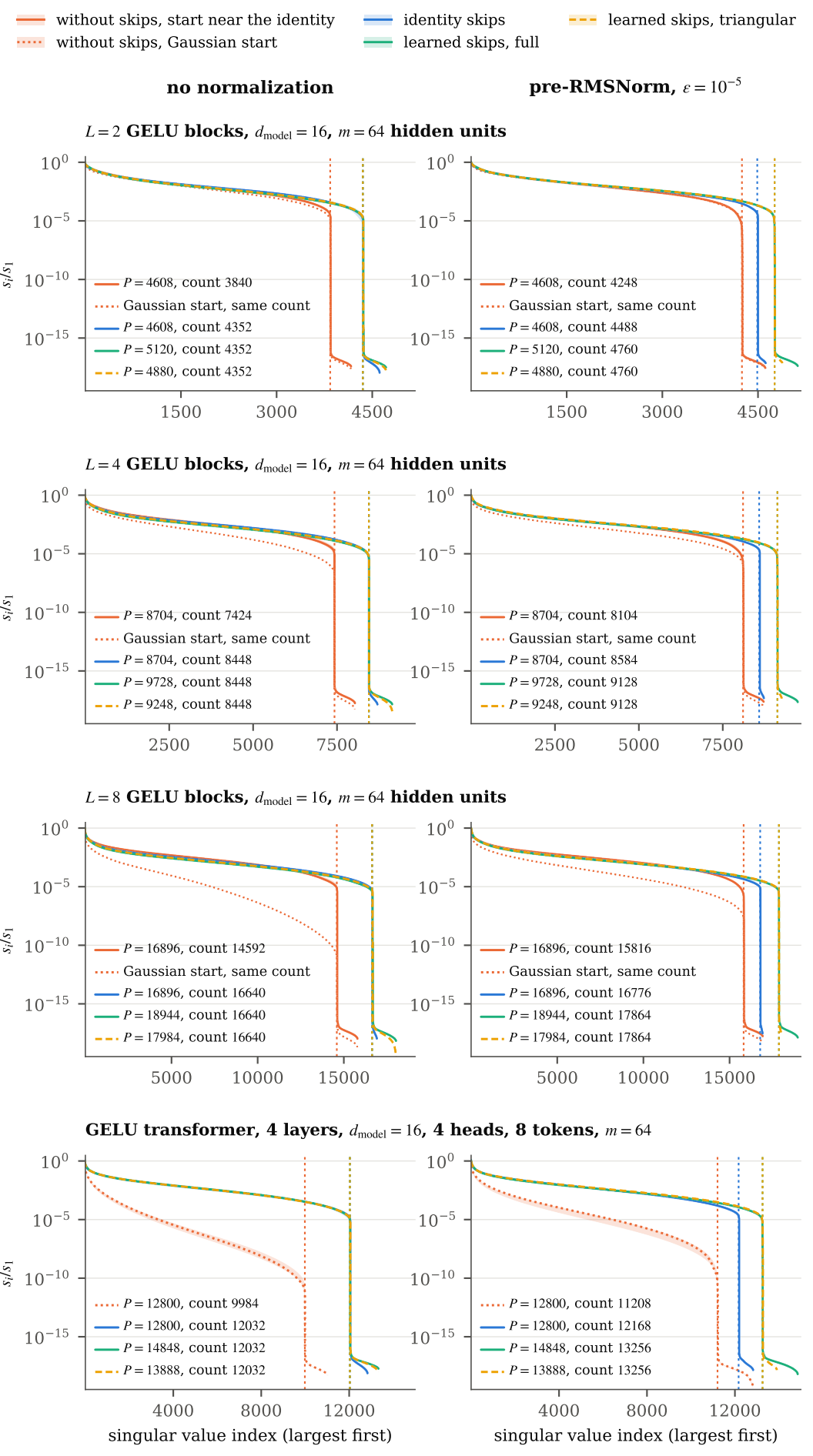
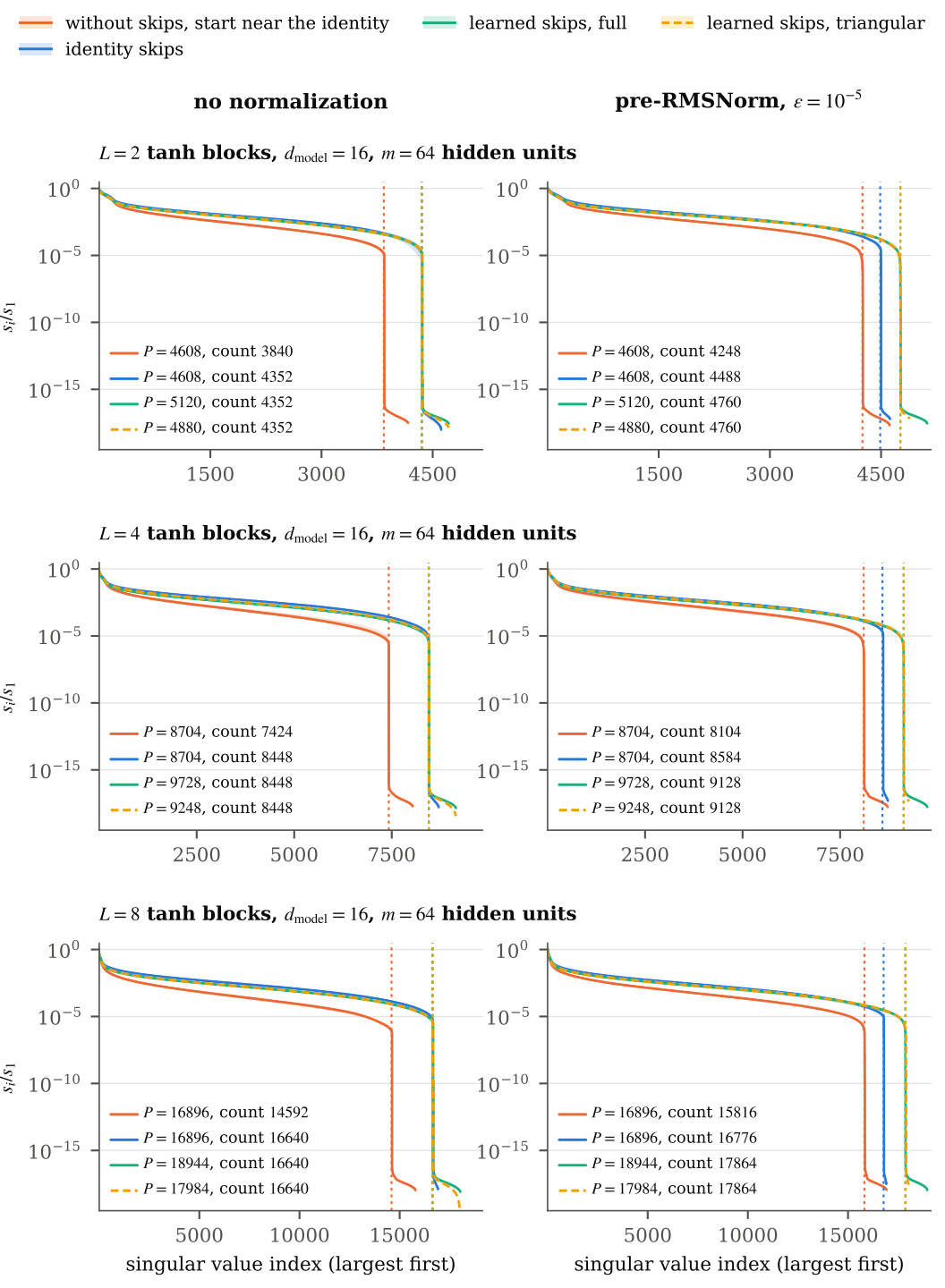
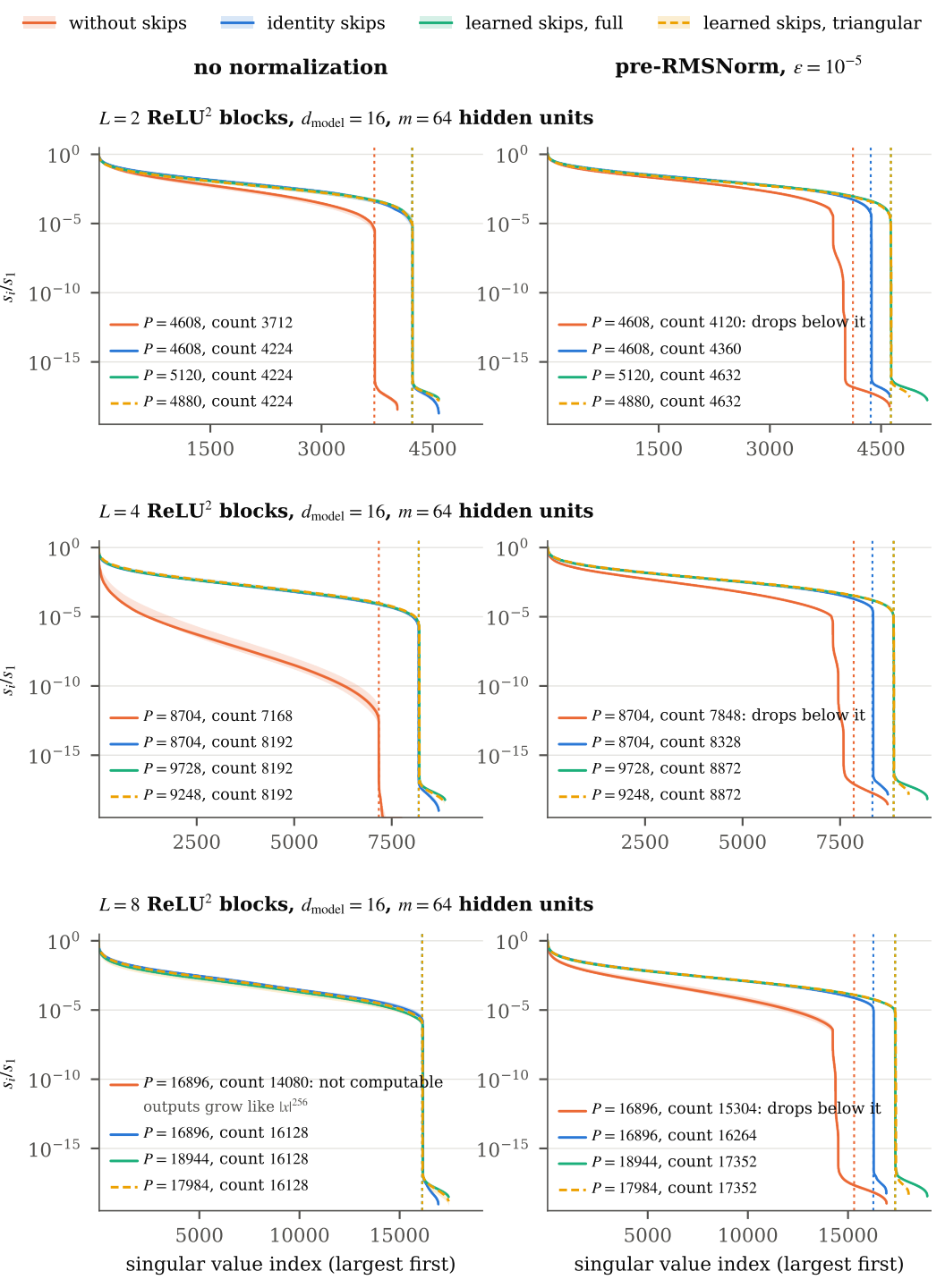
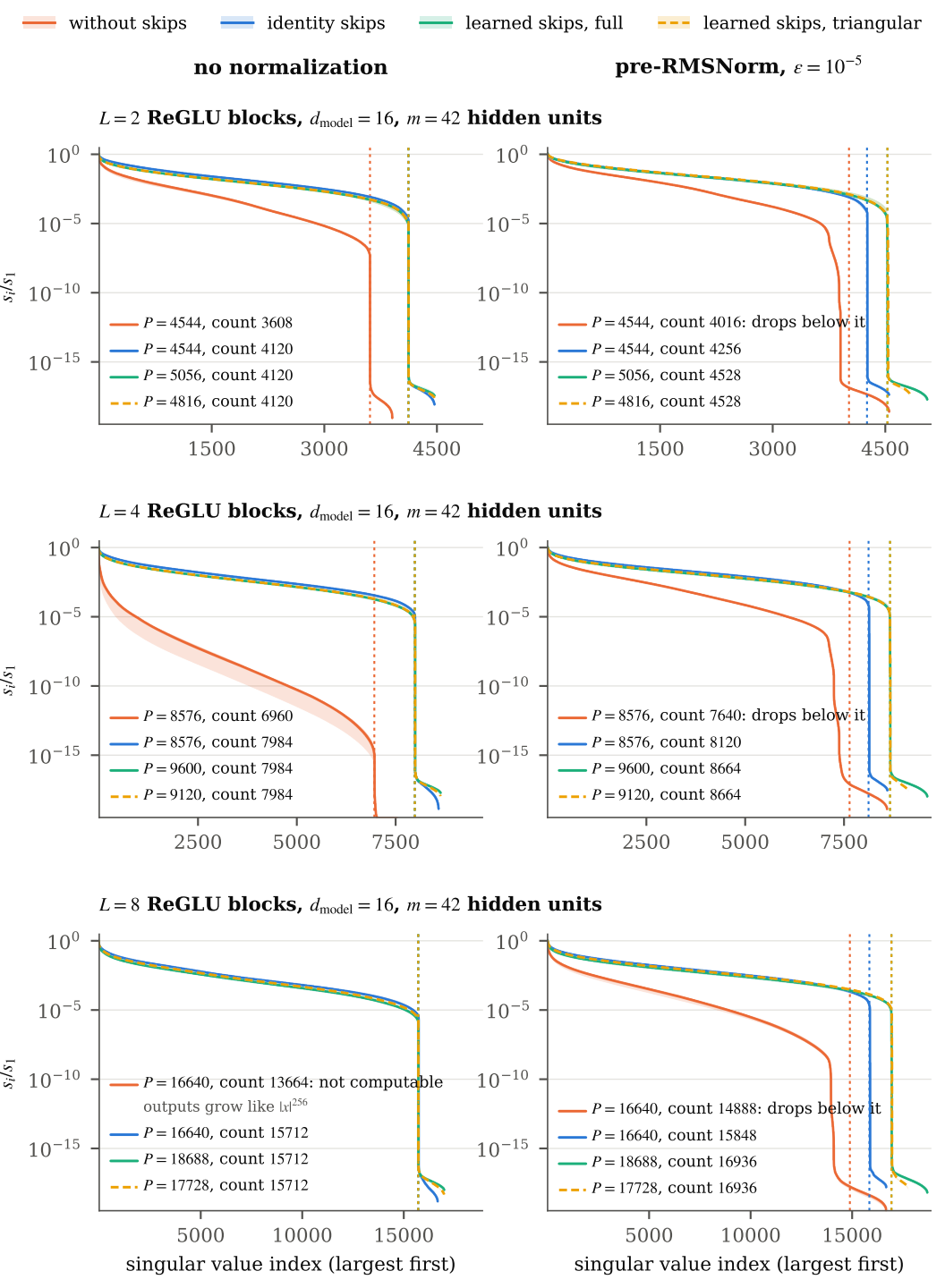
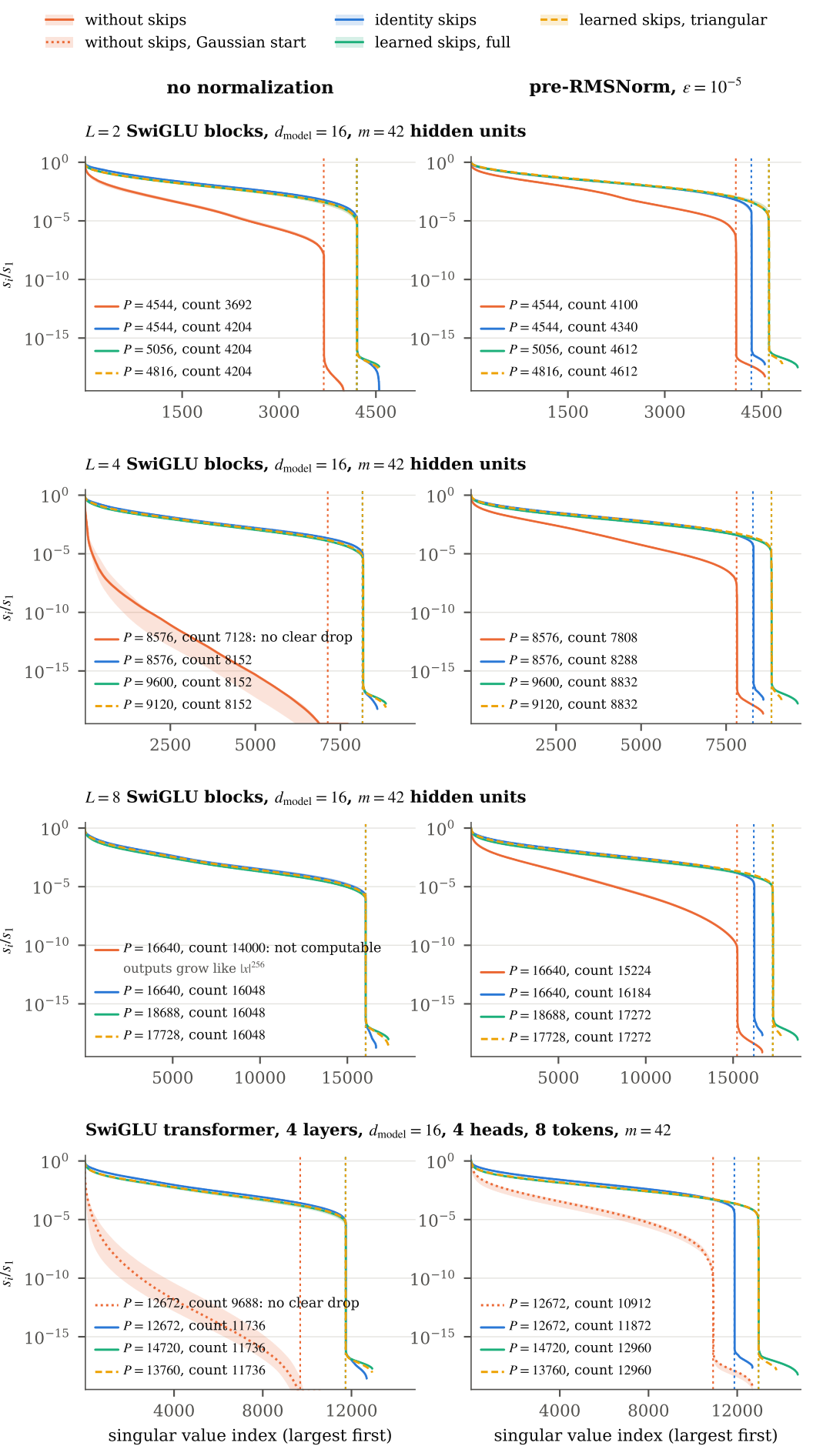

without skips, start near the identity
identity skips
learned skips, full
learned skips, triangular
no normalization
pre-RMSNorm, $\varepsilon = 10^{-5}$
$L = 2$ ReLU blocks, $d_{\text{model}} = 16$, $m = 64$ hidden units
s_i/s_1
10^{0}
10^{-5}
10^{-10}
10^{-15}
$P = 4608$, count 3712
$P = 4608$, count 4224
$P = 5120$, count 4224
$P = 4880$, count 4224
1500
3000
4500
$P = 4608$, count 4120: drops below it
$P = 4608$, count 4360
$P = 5120$, count 4632
$P = 4880$, count 4632
$L = 4$ ReLU blocks, $d_{\text{model}} = 16$, $m = 64$ hidden units
$P = 8704$, count 7168
$P = 8704$, count 8192
$P = 9728$, count 8192
$P = 9248$, count 8192
2500
5000
7500
$P = 8704$, count 7848: drops below it
$P = 8704$, count 8328
$P = 9728$, count 8872
$P = 9248$, count 8872
$L = 8$ ReLU blocks, $d_{\text{model}} = 16$, $m = 64$ hidden units
$P = 16896$, count 14080
$P = 16896$, count 16128
$P = 18944$, count 16128
$P = 17984$, count 16128
5000
10000
15000
singular value index (largest first)
$P = 16896$, count 15304: drops below it
$P = 16896$, count 16264
$P = 18944$, count 17352
$P = 17984$, count 17352
singular value index (largest first)

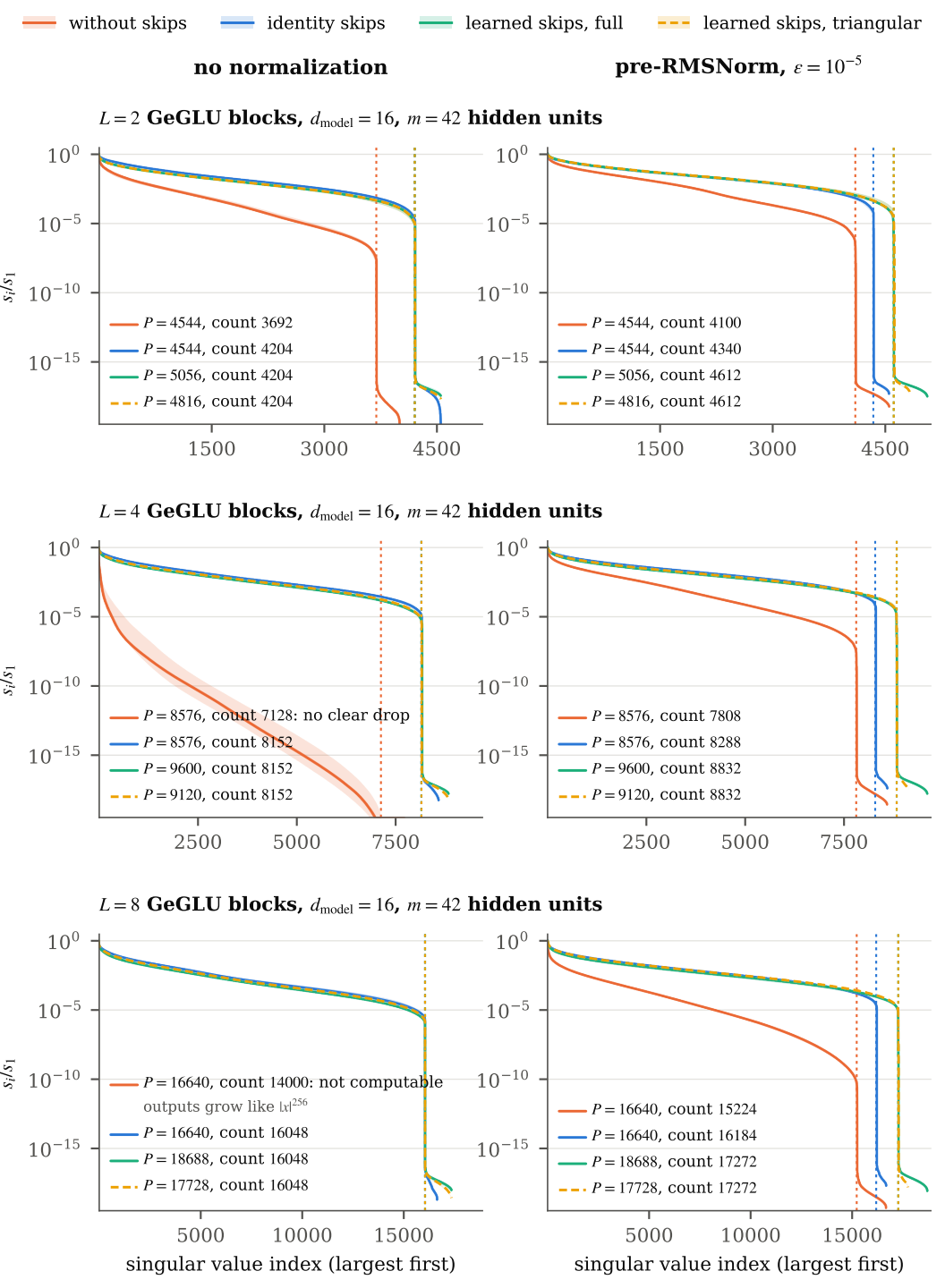

without skips, start near the identity
without skips, Gaussian start
identity skips
learned skips, full
learned skips, triangular
no normalization
pre-RMSNorm, $\varepsilon = 10^{-5}$
$L = 2$ GELU blocks, $d_{\text{model}} = 16$, $m = 64$ hidden units
$L = 4$ GELU blocks, $d_{\text{model}} = 16$, $m = 64$ hidden units
$L = 8$ GELU blocks, $d_{\text{model}} = 16$, $m = 64$ hidden units
GELU transformer, 4 layers, $d_{\text{model}} = 16$, 4 heads, 8 tokens, $m = 64$
s_i/s_1
10^{0}
10^{-5}
10^{-10}
10^{-15}
$P = 4608$, count 3840
Gaussian start, same count
$P = 4608$, count 4352
$P = 5120$, count 4352
$P = 4880$, count 4352
$P = 4608$, count 4248
Gaussian start, same count
$P = 4608$, count 4488
$P = 5120$, count 4760
$P = 4880$, count 4760
1500
3000
4500
$P = 8704$, count 7424
Gaussian start, same count
$P = 8704$, count 8448
$P = 9728$, count 8448
$P = 9248$, count 8448
$P = 8704$, count 8104
Gaussian start, same count
$P = 8704$, count 8584
$P = 9728$, count 9128
$P = 9248$, count 9128
2500
5000
7500
$P = 16896$, count 14592
Gaussian start, same count
$P = 16896$, count 16640
$P = 18944$, count 16640
$P = 17984$, count 16640
$P = 16896$, count 15816
Gaussian start, same count
$P = 16896$, count 16776
$P = 18944$, count 17864
$P = 17984$, count 17864
5000
10000
15000
$P = 12800$, count 9984
$P = 12800$, count 12032
$P = 14848$, count 12032
$P = 13888$, count 12032
$P = 12800$, count 11208
$P = 12800$, count 12168
$P = 14848$, count 13256
$P = 13888$, count 13256
4000
8000
12000
singular value index (largest first)
singular value index (largest first)

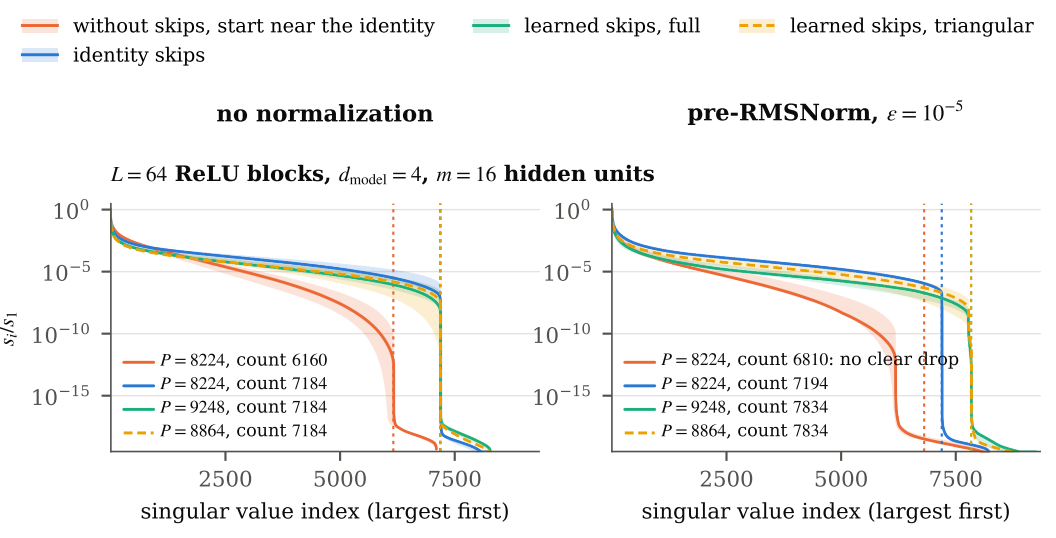

without skips, start near the identity
identity skips
learned skips, full
learned skips, triangular
no normalization
pre-RMSNorm, $\varepsilon = 10^{-5}$
$L = 2$ tanh blocks, $d_{\text{model}} = 16$, $m = 64$ hidden units
$L = 4$ tanh blocks, $d_{\text{model}} = 16$, $m = 64$ hidden units
$L = 8$ tanh blocks, $d_{\text{model}} = 16$, $m = 64$ hidden units
s_i/s_1
10^{0}
10^{-5}
10^{-10}
10^{-15}
$P = 4608$, count 3840
$P = 4608$, count 4352
$P = 5120$, count 4352
$P = 4880$, count 4352
$P = 4608$, count 4248
$P = 4608$, count 4488
$P = 5120$, count 4760
$P = 4880$, count 4760
$P = 8704$, count 7424
$P = 8704$, count 8448
$P = 9728$, count 8448
$P = 9248$, count 8448
$P = 8704$, count 8104
$P = 8704$, count 8584
$P = 9728$, count 9128
$P = 9248$, count 9128
$P = 16896$, count 14592
$P = 16896$, count 16640
$P = 18944$, count 16640
$P = 17984$, count 16640
$P = 16896$, count 15816
$P = 16896$, count 16776
$P = 18944$, count 17864
$P = 17984$, count 17864
1500
3000
4500
2500
5000
7500
10000
15000
singular value index (largest first)
singular value index (largest first)

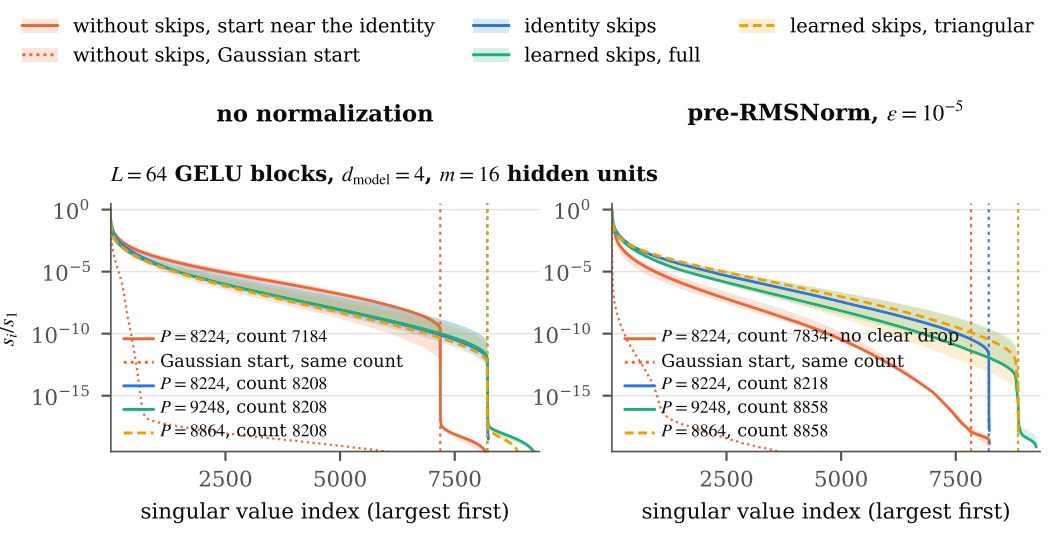

without skips
identity skips
learned skips, full
learned skips, triangular
no normalization
pre-RMSNorm, $\varepsilon = 10^{-5}$
$L = 2$ ReLU^2 blocks, $d_{\text{model}} = 16$, $m = 64$ hidden units
$L = 4$ ReLU^2 blocks, $d_{\text{model}} = 16$, $m = 64$ hidden units
$L = 8$ ReLU^2 blocks, $d_{\text{model}} = 16$, $m = 64$ hidden units
s_i/s_1
10^0
10^{-5}
10^{-10}
10^{-15}
$P = 4608$, count 3712
$P = 4608$, count 4224
$P = 5120$, count 4224
$P = 4880$, count 4224
$P = 4608$, count 4120: drops below it
$P = 4608$, count 4360
$P = 5120$, count 4632
$P = 4880$, count 4632
$P = 8704$, count 7168
$P = 8704$, count 8192
$P = 9728$, count 8192
$P = 9248$, count 8192
$P = 8704$, count 7848: drops below it
$P = 8704$, count 8328
$P = 9728$, count 8872
$P = 9248$, count 8872
$P = 16896$, count 14080: not computable
outputs grow like $|x|^{256}$
$P = 16896$, count 16128
$P = 18944$, count 16128
$P = 17984$, count 16128
$P = 16896$, count 15304: drops below it
$P = 16896$, count 16264
$P = 18944$, count 17352
$P = 17984$, count 17352
1500
3000
4500
2500
5000
7500
10000
15000
singular value index (largest first)

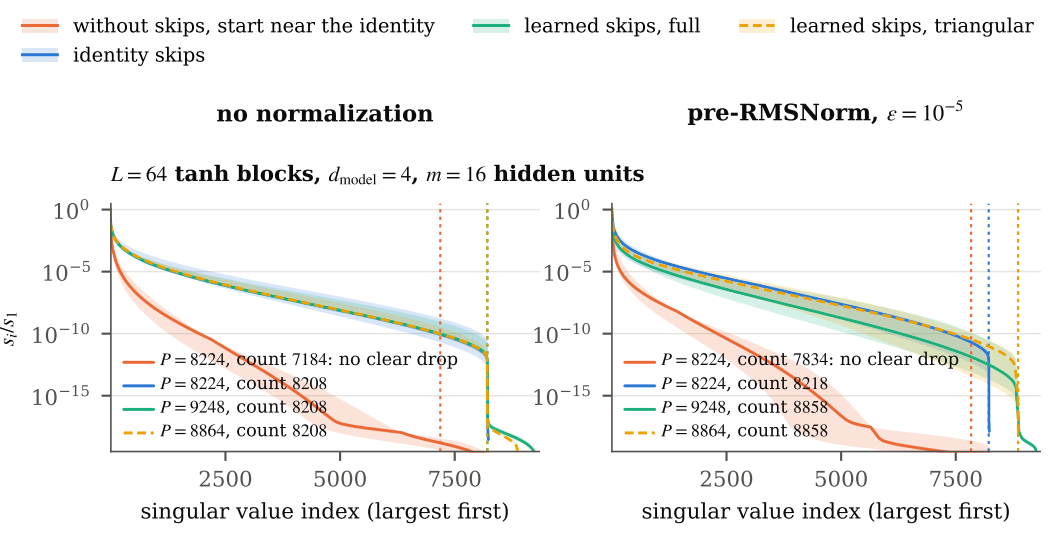

without skips
identity skips
learned skips, full
learned skips, triangular
no normalization
pre-RMSNorm, $\varepsilon = 10^{-5}$
$L = 2$ ReGLU blocks, $d_{\text{model}} = 16$, $m = 42$ hidden units
$L = 4$ ReGLU blocks, $d_{\text{model}} = 16$, $m = 42$ hidden units
$L = 8$ ReGLU blocks, $d_{\text{model}} = 16$, $m = 42$ hidden units
s_i/s_1
10^{0}
10^{-5}
10^{-10}
10^{-15}
$P = 4544$, count 3608
$P = 4544$, count 4120
$P = 5056$, count 4120
$P = 4816$, count 4120
$P = 4544$, count 4016: drops below it
$P = 4544$, count 4256
$P = 5056$, count 4528
$P = 4816$, count 4528
$P = 8576$, count 6960
$P = 8576$, count 7984
$P = 9600$, count 7984
$P = 9120$, count 7984
$P = 8576$, count 7640: drops below it
$P = 8576$, count 8120
$P = 9600$, count 8664
$P = 9120$, count 8664
$P = 16640$, count 13664: not computable
outputs grow like $|x|^{256}$
$P = 16640$, count 15712
$P = 18688$, count 15712
$P = 17728$, count 15712
$P = 16640$, count 14888: drops below it
$P = 16640$, count 15848
$P = 18688$, count 16936
$P = 17728$, count 16936
1500
3000
4500
2500
5000
7500
10000
15000
singular value index (largest first)

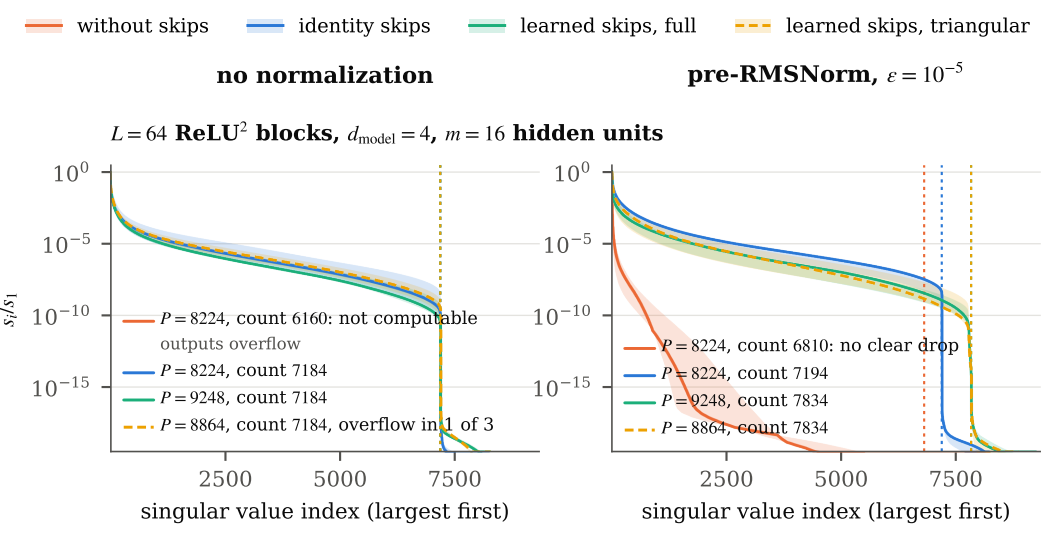

without skips
without skips, Gaussian start
identity skips
learned skips, full
learned skips, triangular
no normalization
pre-RMSNorm, $\varepsilon = 10^{-5}$
$L = 2$ SwiGLU blocks, $d_{\text{model}} = 16$, $m = 42$ hidden units
$P = 4544$, count 3692
$P = 4544$, count 4204
$P = 5056$, count 4204
$P = 4816$, count 4204
$P = 4544$, count 4100
$P = 4544$, count 4340
$P = 5056$, count 4612
$P = 4816$, count 4612
$L = 4$ SwiGLU blocks, $d_{\text{model}} = 16$, $m = 42$ hidden units
$P = 8576$, count 7128: no clear drop
$P = 8576$, count 8152
$P = 9600$, count 8152
$P = 9120$, count 8152
$P = 8576$, count 7808
$P = 8576$, count 8288
$P = 9600$, count 8832
$P = 9120$, count 8832
$L = 8$ SwiGLU blocks, $d_{\text{model}} = 16$, $m = 42$ hidden units
$P = 16640$, count 14000: not computable
outputs grow like $|x|^{256}$
$P = 16640$, count 16048
$P = 18688$, count 16048
$P = 17728$, count 16048
$P = 16640$, count 15224
$P = 16640$, count 16184
$P = 18688$, count 17272
$P = 17728$, count 17272
SwiGLU transformer, 4 layers, $d_{\text{model}} = 16$, 4 heads, 8 tokens, $m = 42$
$P = 12672$, count 9688: no clear drop
$P = 12672$, count 11736
$P = 14720$, count 11736
$P = 13760$, count 11736
$P = 12672$, count 10912
$P = 12672$, count 11872
$P = 14720$, count 12960
$P = 13760$, count 12960
s_i/s_1
singular value index (largest first)

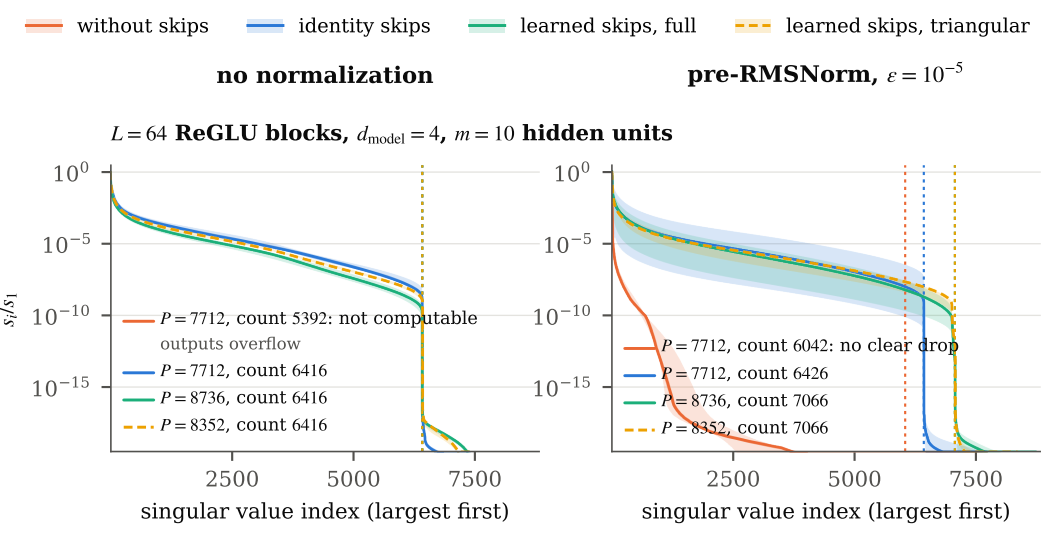

without skips
identity skips
learned skips, full
learned skips, triangular
no normalization
pre-RMSNorm, $\varepsilon = 10^{-5}$
$L = 2$ GeGLU blocks, $d_{\text{model}} = 16$, $m = 42$ hidden units
$L = 4$ GeGLU blocks, $d_{\text{model}} = 16$, $m = 42$ hidden units
$L = 8$ GeGLU blocks, $d_{\text{model}} = 16$, $m = 42$ hidden units
s_i/s_1
singular value index (largest first)
$P = 4544$, count 3692
$P = 4544$, count 4204
$P = 5056$, count 4204
$P = 4816$, count 4204
$P = 4544$, count 4100
$P = 4544$, count 4340
$P = 5056$, count 4612
$P = 4816$, count 4612
$P = 8576$, count 7128: no clear drop
$P = 8576$, count 8152
$P = 9600$, count 8152
$P = 9120$, count 8152
$P = 8576$, count 7808
$P = 8576$, count 8288
$P = 9600$, count 8832
$P = 9120$, count 8832
$P = 16640$, count 14000: not computable
outputs grow like $|x|^{256}$
$P = 16640$, count 16048
$P = 18688$, count 16048
$P = 17728$, count 16048
$P = 16640$, count 15224
$P = 16640$, count 16184
$P = 18688$, count 17272
$P = 17728$, count 17272

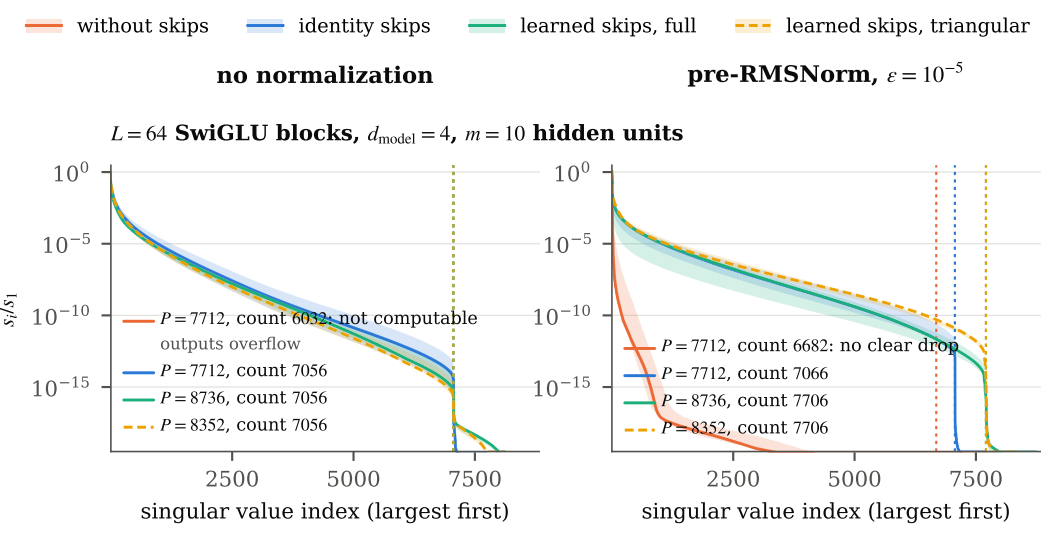

without skips, start near the identity
identity skips
learned skips, full
learned skips, triangular
no normalization
pre-RMSNorm, $\varepsilon = 10^{-5}$
$L = 64$ ReLU blocks, $d_{\text{model}} = 4$, $m = 16$ hidden units
s_i/s_1
10^0
10^{-5}
10^{-10}
10^{-15}
$P = 8224$, count 6160
$P = 8224$, count 7184
$P = 9248$, count 7184
$P = 8864$, count 7184
2500
5000
7500
singular value index (largest first)
$P = 8224$, count 6810: no clear drop
$P = 8224$, count 7194
$P = 9248$, count 7834
$P = 8864$, count 7834

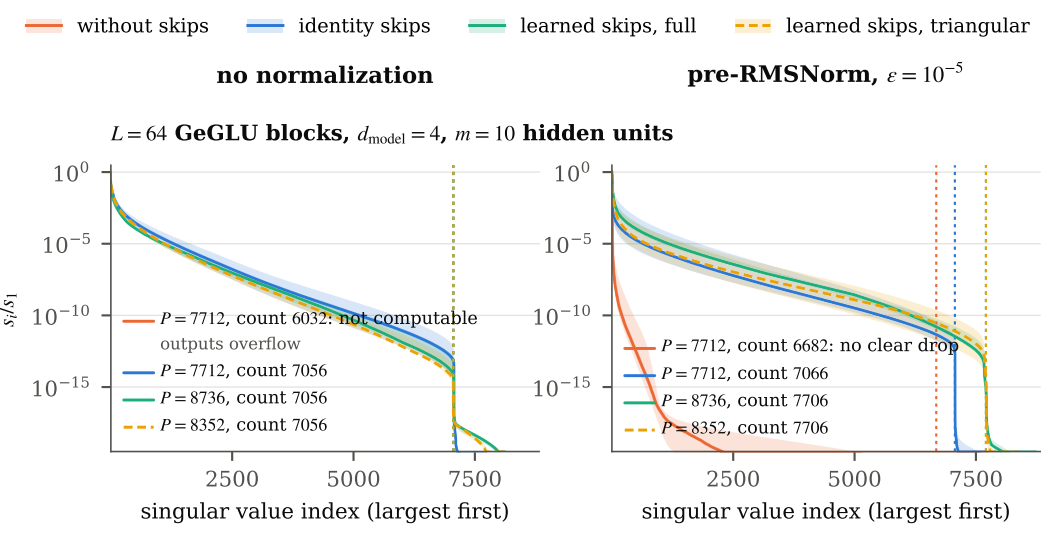

without skips, start near the identity
without skips, Gaussian start
identity skips
learned skips, full
learned skips, triangular
no normalization
pre-RMSNorm, $\varepsilon = 10^{-5}$
$L = 64$ GELU blocks, $d_{\text{model}} = 4$, $m = 16$ hidden units
s_i/s_1
10^{0}
10^{-5}
10^{-10}
10^{-15}
$P = 8224$, count 7184
Gaussian start, same count
$P = 8224$, count 8208
$P = 9248$, count 8208
$P = 8864$, count 8208
$P = 8224$, count 7834: no clear drop
Gaussian start, same count
$P = 8224$, count 8218
$P = 9248$, count 8858
$P = 8864$, count 8858
2500
5000
7500
singular value index (largest first)
singular value index (largest first)

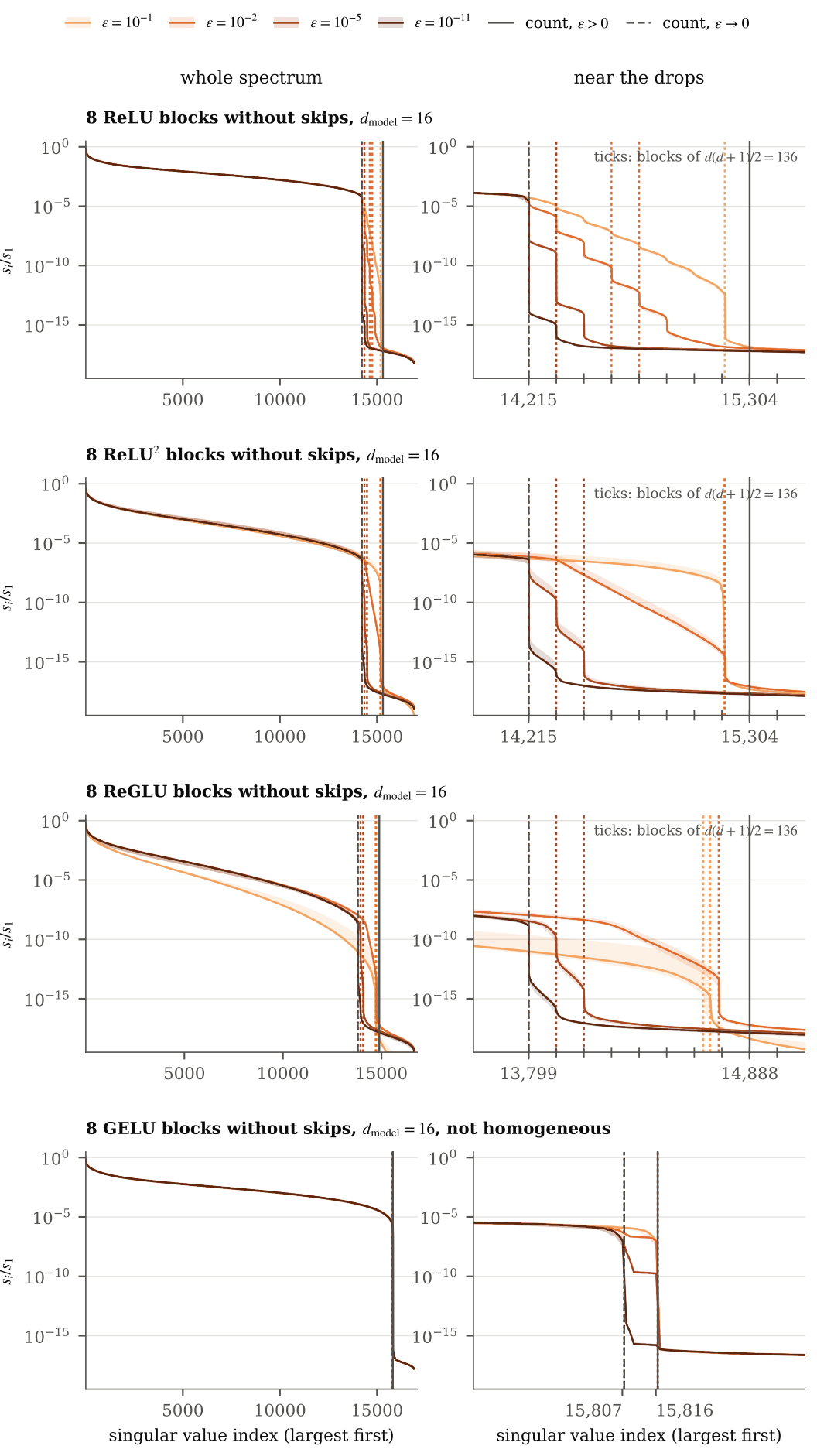

without skips, start near the identity
identity skips
learned skips, full
learned skips, triangular
no normalization
pre-RMSNorm, $\varepsilon = 10^{-5}$
$L = 64$ tanh blocks, $d_{\text{model}} = 4$, $m = 16$ hidden units
s_i/s_1
10^{0}
10^{-5}
10^{-10}
10^{-15}
$P = 8224$, count 7184: no clear drop
$P = 8224$, count 8208
$P = 9248$, count 8208
$P = 8864$, count 8208
$P = 8224$, count 7834: no clear drop
$P = 8224$, count 8218
$P = 9248$, count 8858
$P = 8864$, count 8858
2500
5000
7500
singular value index (largest first)
singular value index (largest first)

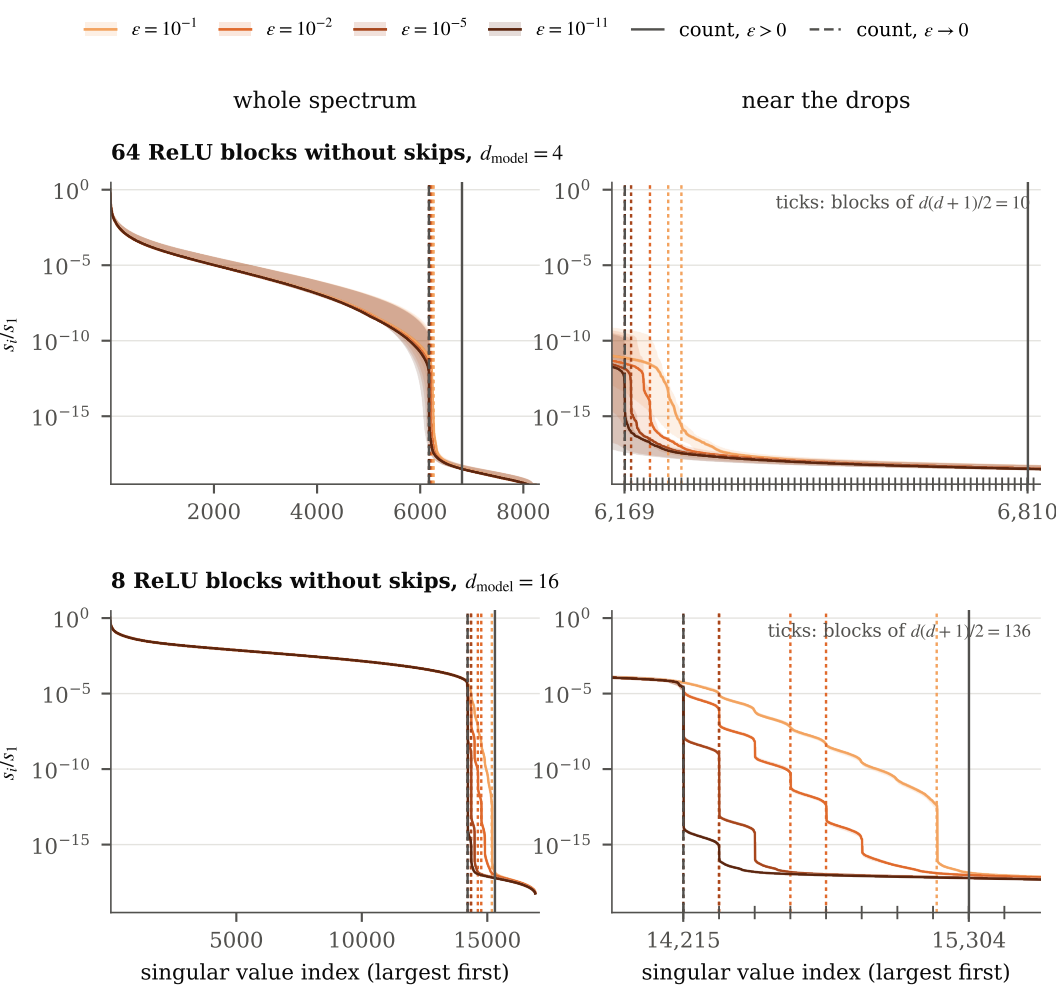

without skips
identity skips
learned skips, full
learned skips, triangular
no normalization
pre-RMSNorm, $\varepsilon = 10^{-5}$
$L = 64$ ReLU^2 blocks, $d_{\text{model}} = 4$, $m = 16$ hidden units
s_i/s_1
10^0
10^{-5}
10^{-10}
10^{-15}
$P = 8224$, count 6160: not computable
outputs overflow
$P = 8224$, count 7184
$P = 9248$, count 7184
$P = 8864$, count 7184, overflow in 1 of 3
$P = 8224$, count 6810: no clear drop
$P = 8224$, count 7194
$P = 9248$, count 7834
$P = 8864$, count 7834
2500
5000
7500
singular value index (largest first)
singular value index (largest first)

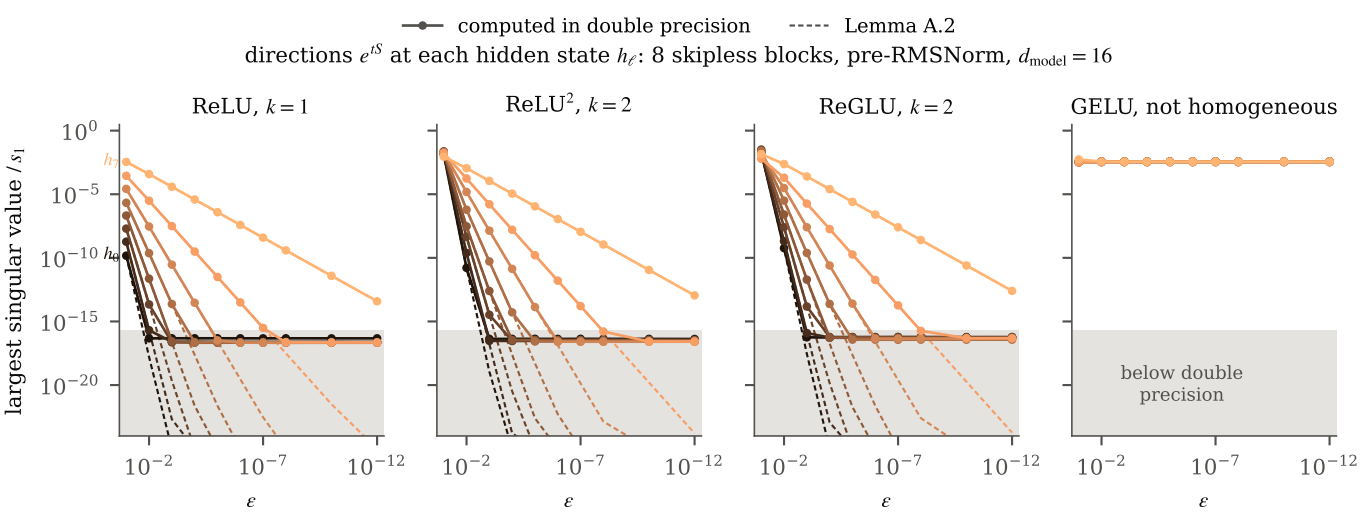

without skips
identity skips
learned skips, full
learned skips, triangular
no normalization
pre-RMSNorm, $\varepsilon = 10^{-5}$
$L = 64$ ReGLU blocks, $d_{\text{model}} = 4$, $m = 10$ hidden units
s_i/s_1
10^{0}
10^{-5}
10^{-10}
10^{-15}
$P = 7712$, count 5392: not computable
outputs overflow
$P = 7712$, count 6416
$P = 8736$, count 6416
$P = 8352$, count 6416
$P = 7712$, count 6042: no clear drop
$P = 7712$, count 6426
$P = 8736$, count 7066
$P = 8352$, count 7066
2500
5000
7500
singular value index (largest first)
singular value index (largest first)


without skips
identity skips
learned skips, full
learned skips, triangular
no normalization
pre-RMSNorm, $\varepsilon = 10^{-5}$
$L = 64$ SwiGLU blocks, $d_{\text{model}} = 4$, $m = 10$ hidden units
s_i/s_1
10^{0}
10^{-5}
10^{-10}
10^{-15}
$P = 7712$, count 6032: not computable
outputs overflow
$P = 7712$, count 7056
$P = 8736$, count 7056
$P = 8352$, count 7056
2500
5000
7500
singular value index (largest first)
$P = 7712$, count 6682: no clear drop
$P = 7712$, count 7066
$P = 8736$, count 7706
$P = 8352$, count 7706
2500
5000
7500
singular value index (largest first)


without skips
identity skips
learned skips, full
learned skips, triangular
no normalization
pre-RMSNorm, $\varepsilon = 10^{-5}$
$L = 64$ GeGLU blocks, $d_{\text{model}} = 4$, $m = 10$ hidden units
s_i/s_1
10^0
10^{-5}
10^{-10}
10^{-15}
$P = 7712$, count 6032: not computable
outputs overflow
$P = 7712$, count 7056
$P = 8736$, count 7056
$P = 8352$, count 7056
2500
5000
7500
singular value index (largest first)
$P = 7712$, count 6682: no clear drop
$P = 7712$, count 7066
$P = 8736$, count 7706
$P = 8352$, count 7706


$\varepsilon = 10^{-1}$
$\varepsilon = 10^{-2}$
$\varepsilon = 10^{-5}$
$\varepsilon = 10^{-11}$
count, $\varepsilon > 0$
count, $\varepsilon \to 0$
whole spectrum
near the drops
8 ReLU blocks without skips, $d_{\text{model}} = 16$
8 ReLU^2 blocks without skips, $d_{\text{model}} = 16$
8 ReGLU blocks without skips, $d_{\text{model}} = 16$
8 GELU blocks without skips, $d_{\text{model}} = 16$, not homogeneous
s_i/s_1
ticks: blocks of $d(d+1)/2 = 136$
14,215
15,304
13,799
14,888
15,807
15,816
5000
10000
15000
singular value index (largest first)


$\varepsilon = 10^{-1}$
$\varepsilon = 10^{-2}$
$\varepsilon = 10^{-5}$
$\varepsilon = 10^{-11}$
count, $\varepsilon > 0$
count, $\varepsilon \to 0$
whole spectrum
near the drops
64 ReLU blocks without skips, $d_{\text{model}} = 4$
s_i/s_1
10^{0}
10^{-5}
10^{-10}
10^{-15}
2000
4000
6000
8000
ticks: blocks of $d(d+1)/2 = 10$
6,169
6,810
8 ReLU blocks without skips, $d_{\text{model}} = 16$
5000
10000
15000
ticks: blocks of $d(d+1)/2 = 136$
14,215
15,304
singular value index (largest first)
singular value index (largest first)


computed in double precision
Lemma A.2
directions e^{tS} at each hidden state h_ℓ: 8 skipless blocks, pre-RMSNorm, $d_{\text{model}} = 16$
ReLU, $k = 1$
ReLU^2, $k = 2$
ReGLU, $k = 2$
GELU, not homogeneous
largest singular value / s_1
10^0
10^{-5}
10^{-10}
10^{-15}
10^{-20}
h_1
h_8
below double precision
10^{-2}
10^{-7}
10^{-12}
ε